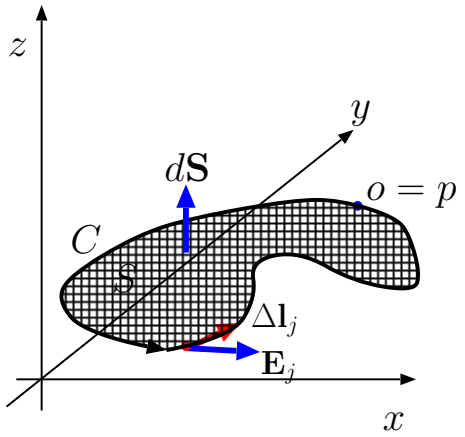
6 Circulation and boundary conditions

Since curl-free static electric fields have path-independent line integrals, it follows that over closed paths C (when points p and o coincide)

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0,$$

where the $\oint_C \mathbf{E} \cdot d\mathbf{l}$ is called the **circulation** of field $\mathbf{E}$ over closed path C bounding a surface S (see margin).



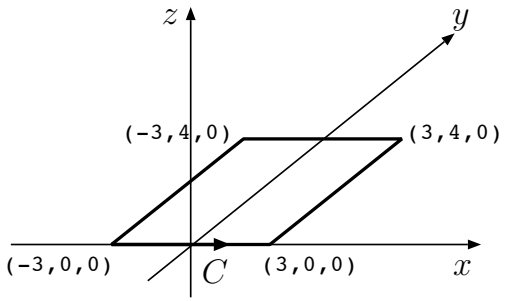
Closed loop integral over path C enclosing surface S.

Note that the area increment dS of surface S is taken by convention to point in the right-hand-rule direction with respect to "circulation" direction C.

Example 1: Consider the static electric field variation

$$\mathbf{E}(x, y, z) = \hat{x}\frac{\rho x}{\epsilon_o}$$

that will be encountered within a uniformly charged slab of an infinite extent in y and z directions and a finite width in x direction centered about $x = 0$. Show that this field $\mathbf{E}$ satisfies the condition $\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$ for a rectangular closed path C with vertices at $(x, y, z) = (-3, 0, 0)$, $(3, 0, 0)$, $(3, 4, 0)$, and $(-3, 4, 0)$ traversed in the order of the vertices given.



Solution: Integration path C is shown in the figure in the margin. With the help of the figure we expand the circulation $\oint_C \mathbf{E} \cdot d\mathbf{l}$ as

$$\begin{aligned}\mathbf{E} &= \int_{x=-3}^{3} \hat{x}\frac{\rho x}{\epsilon_o} \cdot \hat{x}dx + \int_{y=0}^{4} \hat{x}\frac{\rho 3}{\epsilon_o} \cdot \hat{y}dy + \int_{x=3}^{-3} \hat{x}\frac{\rho x}{\epsilon_o} \cdot \hat{x}dx + \int_{y=4}^{0} \hat{x}\frac{\rho(-3)}{\epsilon_o} \cdot \hat{y}dy \\ &= \int_{x=-3}^{3} \frac{\rho x}{\epsilon_o}dx + 0 + \int_{x=3}^{-3} \frac{\rho x}{\epsilon_o}dx + 0 = 0.\end{aligned}$$

Note that in expanding $\oint_C \mathbf{E} \cdot d\mathbf{l}$ above for the given path C, we took $d\mathbf{l}$ as $\hat{x}dx$ and $\hat{y}dy$ in turns (along horizontal and vertical edges of C, respectively) and ordered the integration limits in x and y to traverse C in a counter-clockwise direction as indicated in the diagram.

- Vector fields $\mathbf{E}$ having zero circulations over all closed paths C are known as **conservative fields** (for obvious reasons having to do with their use in modeling static fields compatible with conservation theorems).

  – The concepts of *curl-free* and *conservative* fields overlap, that is

  $$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0 \quad \Leftrightarrow \quad \nabla \times \mathbf{E} = 0$$

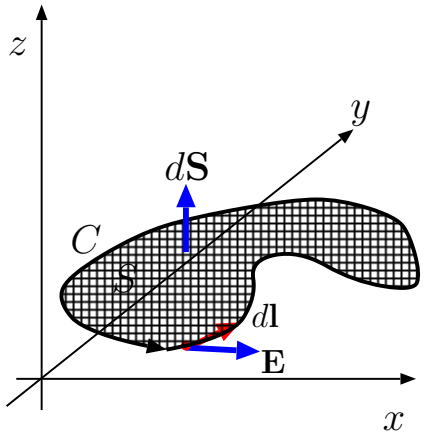
  over all closed paths C and at each $\mathbf{r}$.

- The above relationship between circulation and curl is also a consequence of **Stoke's theorem** (discussed in MATH 241) which asserts that

  $$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{E} \cdot d\mathbf{S},$$

  where

  – the integration surface S on the right is bounded by the closed integration contour C of the left side, and



STOKE'S THM:
Circulation of E around close path C equals the flux over enclosed surface S of the curl of E taken in direction of dS.

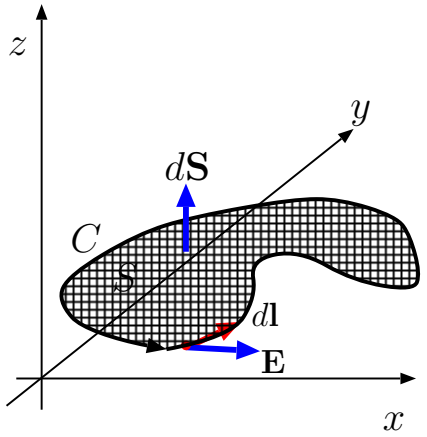
dS points in right-hand-rule direction with respect to "circulation" direction C.

Stoke's thm.

– the incremental area element $d\mathbf{S}$ on the right points across area S in the direction indicated by a **right-hand rule** as follows:

Point your right thumb in chosen circulation direction C; then your right fingers point through surface S in the direction that should be adopted for $d\mathbf{S}$.

– Given Stoke's theorem, $\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$ follows immediately for all C, if $\nabla \times \mathbf{E} = 0$ is true over all $\mathbf{r}$.



STOKE'S THM:
Circulation of E around close path C equals the flux over enclosed surface S of the curl of E taken in direction of dS.

dS points in right-hand-rule direction with respect to "circulation" direction C.

Verification of Stoke's thm: Stoke's theorem applies to all contours C of all sizes and orientations and their enclosed surfaces S of any shape. For a small rectangular contour on a constant x plane with sufficiently small Δy and Δz dimensions parallel to y and z axes (see figure in the margin), we have

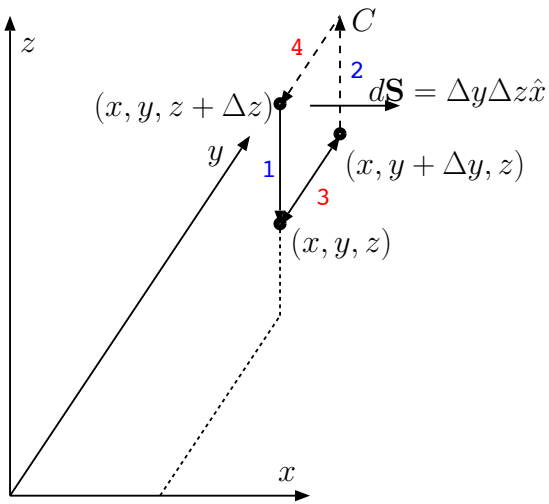
$$\oint_C \mathbf{E} \cdot d\mathbf{l} \approx (E_{z|2} - E_{z|1})\Delta z - (E_{y|4} - E_{y|3})\Delta y,$$

an approximation that can also be re-arranged as

$$\oint_C \mathbf{E} \cdot d\mathbf{l} \approx (\frac{E_{z|2} - E_{z|1}}{\Delta y} - \frac{E_{y|4} - E_{y|3}}{\Delta z})\hat{x} \cdot \Delta y \Delta z\, \hat{x}.$$

Right hand side above is clearly an approximation also for

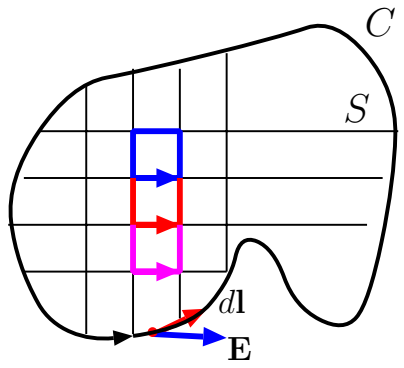
$$(\nabla \times \mathbf{E}) \cdot d\mathbf{S} = (\nabla \times \mathbf{E}) \cdot\, dydz\, \hat{x} = (\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z})\hat{x}\, \cdot dydz\hat{x}.$$



Matching the approximations and taking the limit of vanishing Δy and Δz, we realize that for any infinitesimal area element $d\mathbf{S}$ of an arbitrary direction,

$$\nabla \times \mathbf{E} \cdot d\mathbf{S} = \oint_{dC} \mathbf{E} \cdot d\mathbf{l}$$

where dC denotes the bounding infinitesimal contour of $d\mathbf{S}$ traversed in the right-hand rule direction. Stokes theorem for an arbitrary C over a finite enclosed area S is obtained by superposing these infinitesimals — the left side then becomes $\int_S \nabla \times \mathbf{E} \cdot d\mathbf{S}$ and the right side $\oint_C \mathbf{E} \cdot d\mathbf{l}$ after cancellations of opposing line integral contributions coming from overlapping adjacent segments (see figure in the margin).



Sum of circulations over small squares cancel in the interior edges and only survive around the exterior path C. This way, circulation around C matches the sum of the fluxes of curl E calculated over the small squares.

  - Stoke's theorem clearly implies that **curl is circulation per unit area**, just as the divergence theorem showed that **divergence is flux per unit volume**. The only difference is, curl also has a direction, which is the normal unit of the plane that contains the maximal value of circulation per unit area found at that location over all possible orientations of $d\mathbf{S}$.

We can now summarize the general constraints governing static electric fields as

$$\nabla \times \mathbf{E}(\mathbf{r}) = 0, \quad \nabla \cdot \mathbf{D}(\mathbf{r}) = \rho(\mathbf{r}), \quad \text{where } \mathbf{D}(\mathbf{r}) = \epsilon_o \mathbf{E}(\mathbf{r}).$$

- Vector fields $\mathbf{E}(\mathbf{r})$ and $\mathbf{D}(\mathbf{r})$ governed by these equations will in general be continuous functions of position coordinates $\mathbf{r} = (x, y, z)$ *except* at

Laws of electrostatics:

$$\nabla \times \mathbf{E} = 0$$
$$\nabla \cdot \epsilon_o \mathbf{E} = \rho$$

They also apply "quasi-statically" over a region of dimension L when a time-varying field source $\rho(\mathbf{r}, t)$ has a *time-constant* τ much longer than the propagation time delay L/c of $\mathbf{E}(\mathbf{r}, t)$ field variations across the region (c is the speed of light).

In electro-quasistatics (EQS) $\mathbf{E}(\mathbf{r}, t)$ will be accompanied by a slowly varying magnetic field $\mathbf{B}(\mathbf{r}, t)$ (to be studied starting in Lecture 12).

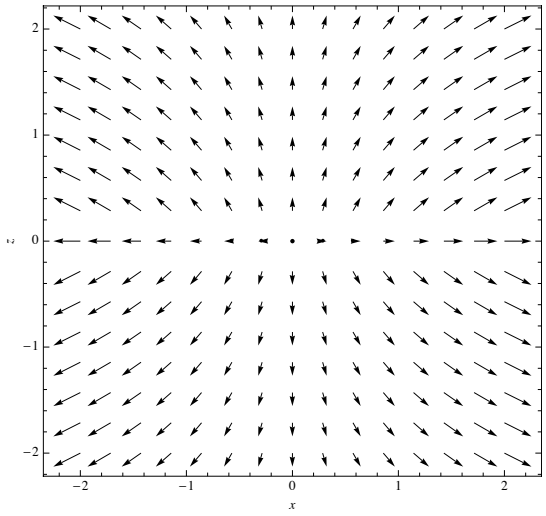
boundary surfaces where charge density function $\rho(\mathbf{r})$ requires a representation in terms of a surface charge density $\rho_s(\mathbf{r})$.

– For instance, according to our earlier results, static electric field of a charge density (see sketch at the margin)

$$\rho(\mathbf{r}) = \rho_s \delta(z)$$

would be

$$\mathbf{E}(\mathbf{r}) = \hat{z}\frac{\rho_s}{2\epsilon_o}\mathrm{sgn}(z) \quad \Rightarrow \quad \mathbf{D}(\mathbf{r}) = \hat{z}\frac{\rho_s}{2}\mathrm{sgn}(z).$$



∘ Consider a superposition of these fields with fields $\mathbf{E}_o(\mathbf{r})$ and $\mathbf{D}_o(\mathbf{r}) = \epsilon_o \mathbf{E}_o(\mathbf{r})$ produced by arbitrary continuous sources, namely (macroscopic) fields

$$\mathbf{E}(\mathbf{r}) = \hat{z}\frac{\rho_s}{2\epsilon_o}\mathrm{sgn}(z) + \mathbf{E}_o(\mathbf{r}) \quad \text{and} \quad \mathbf{D}(\mathbf{r}) = \hat{z}\frac{\rho_s}{2}\mathrm{sgn}(z) + \epsilon_o \mathbf{E}_o(\mathbf{r}).$$

Since fields $\mathbf{E}_o(\mathbf{r})$ and $\mathbf{D}_o(\mathbf{r})$ vary continuously, these field expressions must satisfy

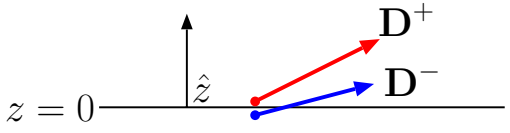
$$\hat{z} \cdot (\mathbf{D}^+ - \mathbf{D}^-) = \rho_s \quad \text{and} \quad \hat{z} \times (\mathbf{E}^+ - \mathbf{E}^-) = 0$$

where

$$\mathbf{E}^+ \equiv \mathbf{E}(x, y, 0^+) \quad \text{and} \quad \mathbf{E}^- \equiv \mathbf{E}(x, y, 0^-)$$

refer to limiting values of $\mathbf{E}$ at $z = 0$ plane from *above* and *below*, respectively, and likewise for

$$\mathbf{D}^+ \equiv \mathbf{D}(x, y, 0^+) \quad \text{and} \quad \mathbf{D}^- \equiv \mathbf{D}(x, y, 0^-).$$



- The above "boundary condition equations" can be written in a more general form (see margin for justification) as

$$\hat{n} \cdot (\mathbf{D}^+ - \mathbf{D}^-) = \rho_s \quad \text{and} \quad \hat{n} \times (\mathbf{E}^+ - \mathbf{E}^-) = 0$$

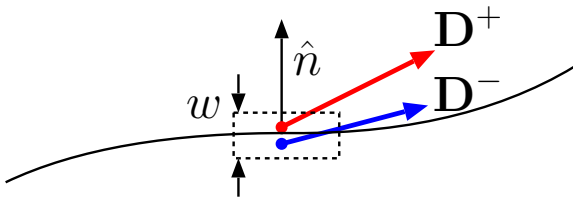
where $\hat{n}$ denotes a unit vector normal to any surface of an arbitrary orientation carrying a surface charge density ρ_s, while field vectors with superscripts $+$ and $-$ indicate limiting values of fields measured on either side of the charged surface (with $\hat{n}$ pointing from $-$ to $+$).

  - The equations can be further simplified as

$$D_n^+ - D_n^- = \rho_s \quad \text{and} \quad E_t^+ = E_t^-$$

    where D_n and E_t refer to normal component of $\mathbf{D}$ and tangential component of $\mathbf{E}$, respectively. Clearly, these **boundary conditions** say that at any surface S,

    - tangential component of electric field $\mathbf{E}$ needs to be continuous, but
    - normal component of $\mathbf{D}$ can change by an amount equal to the charge density ρ_s carried by the surface.



Constraint

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$$

around the dotted path yields

$$E_t^+ = E_t^-$$

in $w \to 0$ limit.

Gauss's law

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_V$$

applied over the dotted volume (seen in profile) yields

$$D_n^+ - D_n^- = \rho_s$$

in $w \to 0$ limit.

Example 2:

Measurements indicate that $\mathbf{D} = 0$ in the region $x < 0$.

Also, $x = 0$ and $x = 5$ m planes contain surface charge densities of $\rho_s = 2$ C/m^2 and ρ_{so}, respectively.

Determine ρ_{so} and $\mathbf{D}$ for $-\infty < x < \infty$ if there are no other charge distributions.

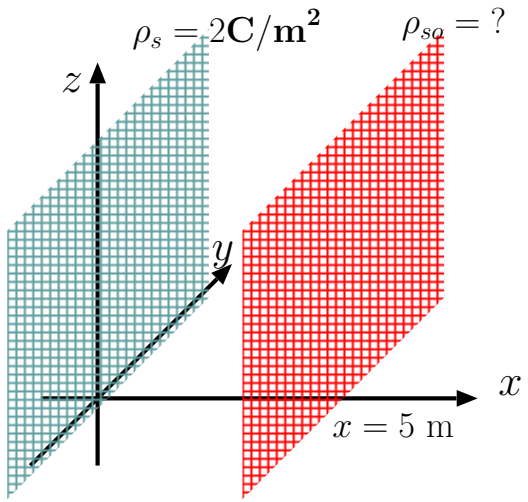
Solution:

Since the normal component of $\mathbf{D}$ must increase by $\rho_s = 2$ C/m^2 when we cross the charged surface $x = 0$, we must have $\mathbf{D} = \hat{x}2$ C/m^2 in the region $0 < x < 5$ m.

Having $\mathbf{D} = 0$ in the region $x < 0$ requires that the field due to surface charge ρ_{so} on $x = 5$ m plane must cancel the field due $\rho_s = 2$ C/m^2 on $x = 0$ plane — this requires that ρ_{so} be -2 C/m^2.

In that case $\mathbf{D} = 0$ in the region $x > 5$ m, because $\mathbf{D}$ must increase by $\rho_{so} = -2$ C/m^2 when we cross the charged surface at $x = 5$ m.

$\mathbf{D} = 0$ for $x < 0$.



Example 3: In the region $x < 0$ measurements indicate a constant displacement field $\mathbf{D} = 3\hat{y}$ C/m^2. Also, $x = 0$ and $x = 5$ m planes contain surface charge densities of $\rho_s = 2$ C/m^2 and $\rho_s = -6$ C/m^2 respectively. Determine $\mathbf{D}$ for $x > 0$ if $\mathbf{D}$ is known to be uniform in the intervals $0 < x < 5$ m and $x > 5$ m.

Solution: First we note that $\mathbf{E} = \frac{\mathbf{D}}{\epsilon_o} = \hat{y}\frac{3}{\epsilon_o}$ V/m is tangential to $x = 0$ and $x = 5$ m surfaces. Since the tangential component of $\mathbf{E}$ cannot change at any boundary, we will have a uniform $E_y = \frac{3}{\epsilon_o}$ in all regions, $-\infty < x < \infty$, implying that $D_y = 3$ C/m^2 throughout (caused by charges at $|y| \to \infty$).

Second, we note that normal component of $\mathbf{D}$ with respect to $x = 0$ and $x = 5$ m surfaces, namely D_x, is zero in $z < 0$. Since the normal component of $\mathbf{D}$ must increase by an amount ρ_s when we cross a charged surface, we must have $D_x = 2$ C/m^2 in the region $0 < x < 5$ m, and $D_x = 2 + (-6) = -4$ C/m^2 in $x > 5$ m.

In summary,

$$\mathbf{D} = \begin{cases} \hat{y}3, & \text{for } x < 0, \\ \hat{x}2 + \hat{y}3, & \text{for } 0 < x < 5 \text{ m} \\ -\hat{x}4 + \hat{y}3, & \text{for } x > 5 \text{ m} \end{cases} \frac{\text{C}}{\text{m}^2}.$$

$\mathbf{D} = 3\hat{y}$ for $x < 0$.

